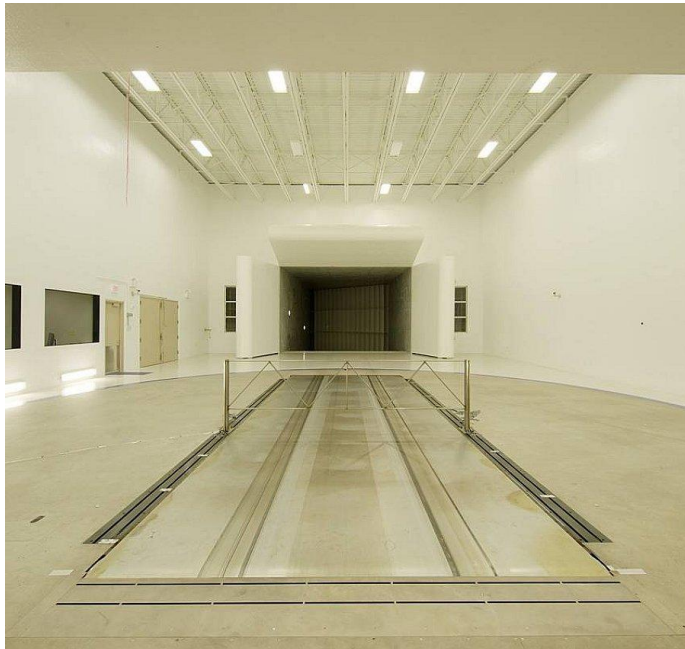
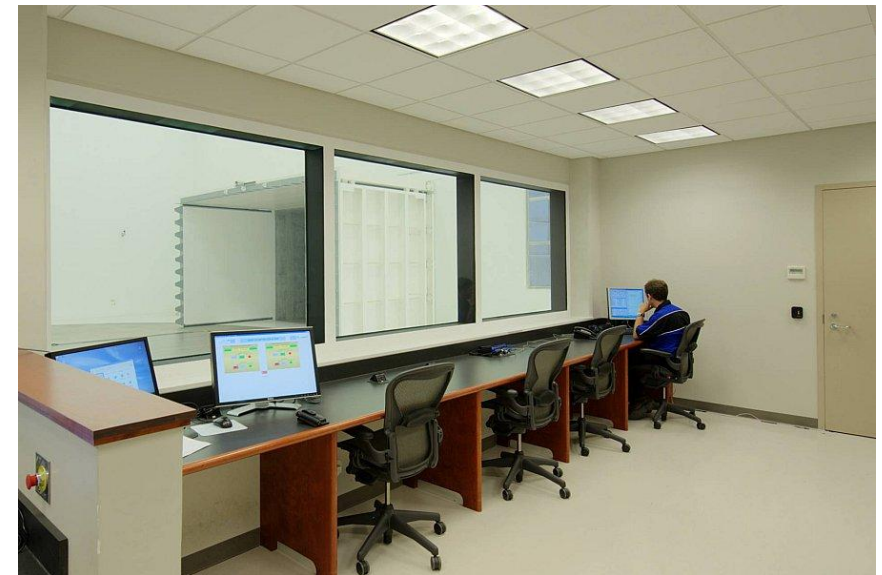


Experimental Methods for Automotive Cooling Module Airflow and Drag Characterization



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Abstract



During the operation of a passenger vehicle, waste heat is generated in the drive unit, electrical systems, and passenger compartment. This heat must be rejected to the ambient air, typically through the use of a cooling module comprising a radiator, condenser, air guide, and fan system. For electric vehicles in particular, efficiency in cooling drag and fan performance can improve range through reduced losses and power consumption. This presentation will describe the methods used for measuring mass air flow and drag of a cooling module in a rolling road wind tunnel.

The module was instrumented with vane anemometers to measure air velocity, and by controlling an adjustable grille shutter, flow to the module was metered. For each data point, the fan motors were operated at a desired rotational speed, and the wind tunnel was set to the road speed of interest. The points were selected from a typical operating range of a passenger vehicle.

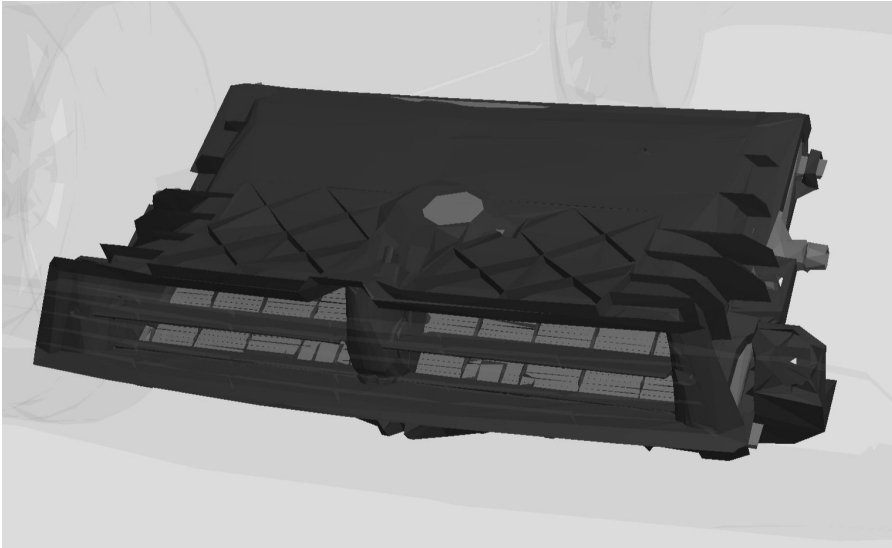
This experimental data was used to characterize the flow volume, uniformity, and drag at typical operational speeds in order to drive design decisions, make correlations to CFD models, and confirm if performance targets were being met.

Aerobuck

- Full scale vehicle model
- Welded rectangular steel tube frame
- Body made of foam and clay
- Some parts were modular to test different configurations
- Used primarily for testing of vehicle aerodynamic drag and high-speed stability
- Airflow tests to understand the effect of cooling system drag and quantify cooling airflow

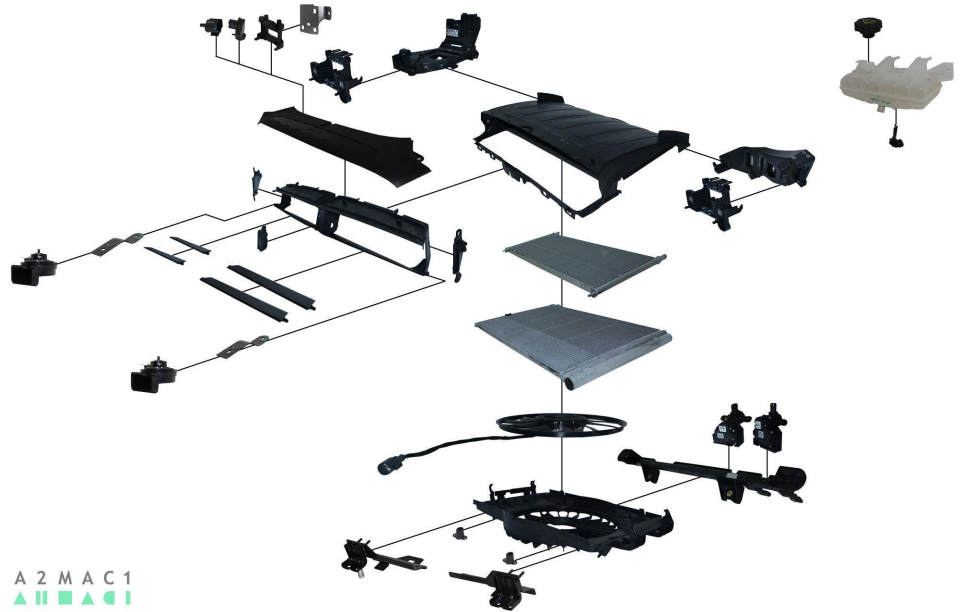
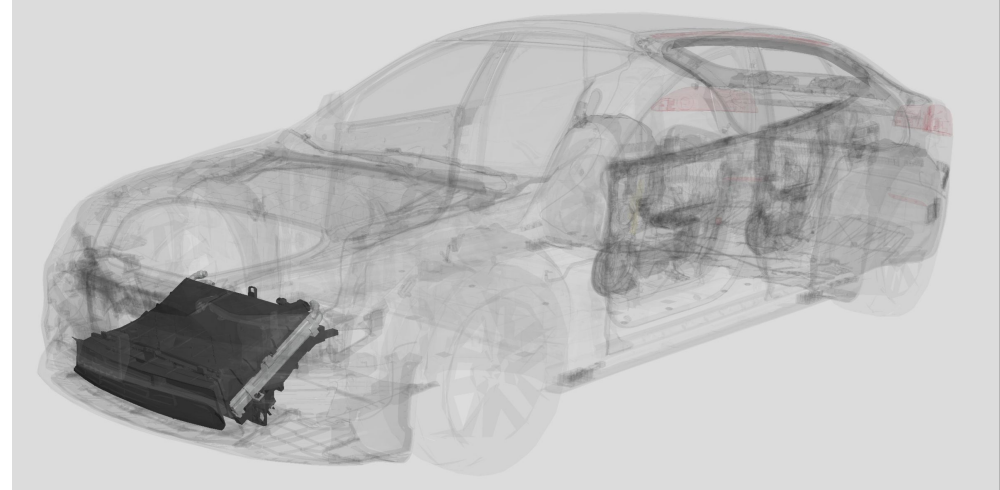


Cooling Pack



Components

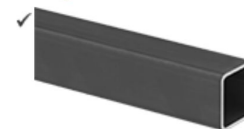
- Air guide (duct)
- Stone Guard
- Radiator
- Condenser
- Fan Shroud
- Fan Blades and Motors
- Mounts and Isolators
- Active Grille Shutter (AGS)



Aerobuck Cooling Pack

- Aerobuck I
 - Welded tube frame
 - Sheet metal fan shroud
 - Representative radiator (no condenser)
 - Carryover fans
 - 3D printed air duct
 - 3D printed AGS (prototype design)
- Aerobuck II
 - Welded tube frame
 - Machined fan shroud and fans
 - Production intent heat exchangers
 - 3D printed air duct
 - 3D printed AGS (prototype design)
- Aerobuck III
 - Prototype cooling pack (production intent)
 - 3D printed stone guard
 - 3D printed air duct
 - 3D printed AGS (production intent)

Easy-to-Weld 4130 Alloy Steel Rectangular Tubes



- Yield Strength: 75,000 psi
- Hardness: Rockwell C20 (Hard)
- Heat Treatable: Yes
- Max. Hardness After Heat Treatment: Rockwell C49
- Specifications Met: SAE AMS-T-6736, MIL-T-6736

4130 alloy steel has a low carbon content for good weldability. It's often used for structural applications.

Aluminum Twist-Resistant Rivet Nuts



These rivet nuts are one-third the weight of steel rivet nuts and mildly corrosion resistant. A ribbed body holds the nut in place so it won't turn when tightening a screw. Also known as blind inserts, they can be installed when you have access to only one side of the material. Install in a drilled hole using a [rivet nut tool](#). The tool collapses the nut, creating a backside flange that holds the nut to the material. Length is measured from under the flange. Installed length is the maximum length of the barrel when installed.

 For technical drawings and 3-D models, click on a part number.



Heat Exchangers

- Radiator and Condenser
- Inlet/Outlet Tanks
- Core
 - Tubes
 - Fins

$$Q = \dot{m}_c c_c \Delta T_c = \dot{m}_a c_a \Delta T_a = UA \Delta T_m$$

Q = heat exchange rate

$\dot{m}_c$ = mass flow rate, coolant

c_c = specific heat, coolant, $3283 \frac{J}{kg \cdot K}$ (50% water, 50% ethylene glycol)

ΔT_c = temperature difference, coolant

$\dot{m}_a$ = mass flow rate, air

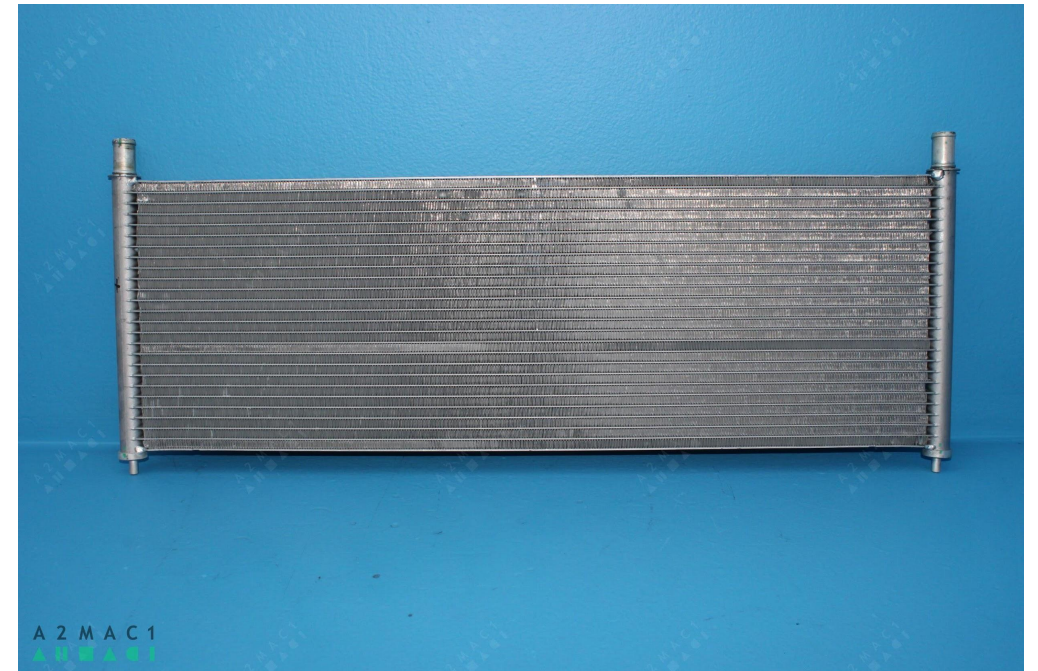
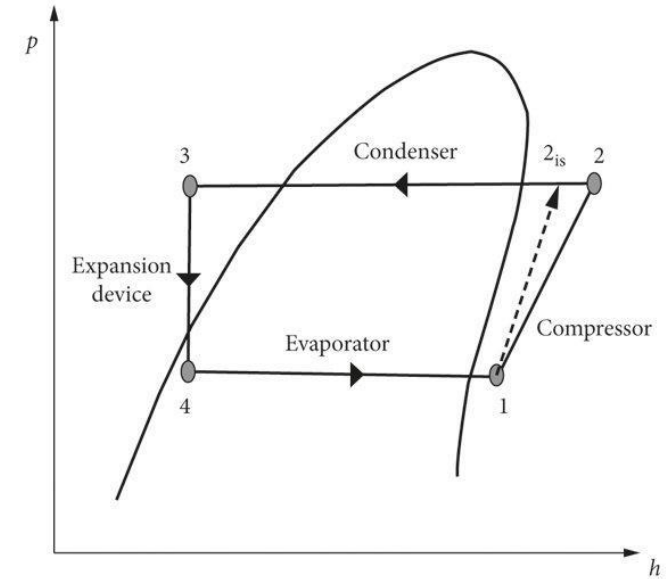
c_a = specific heat, air, $1000 \frac{J}{kg \cdot K}$

ΔT_a = temperature difference, air

U = overall heat transfer coefficient

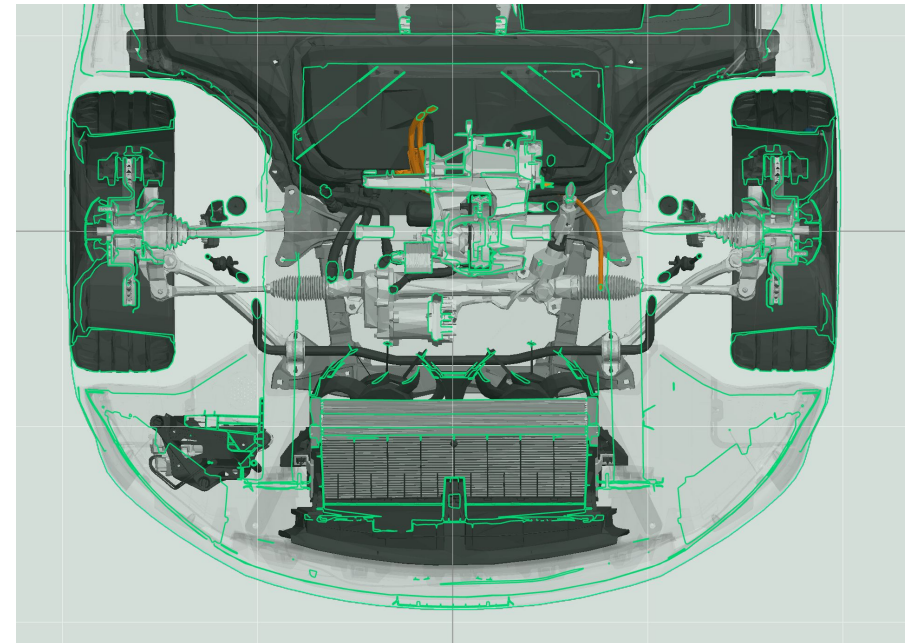
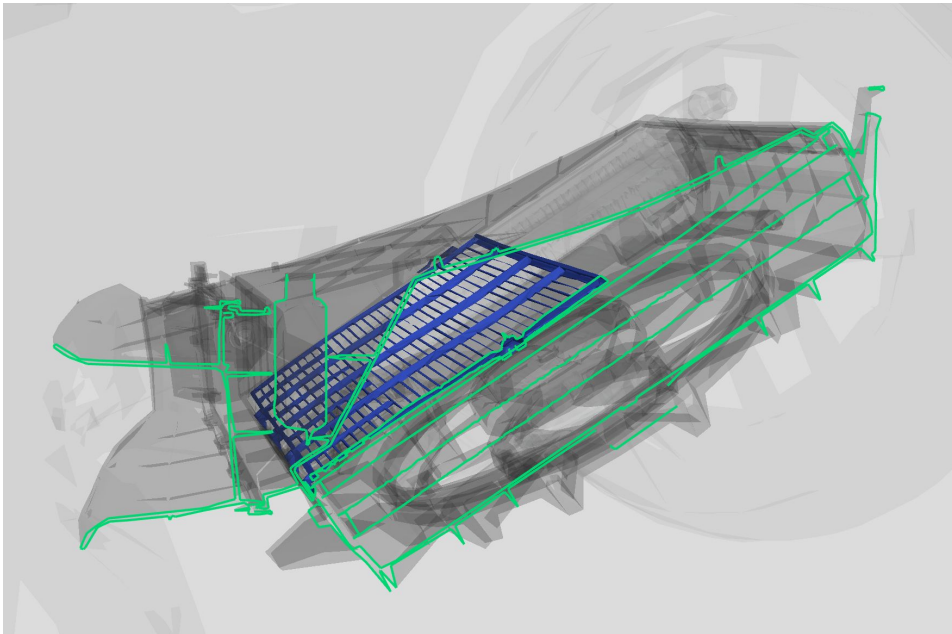
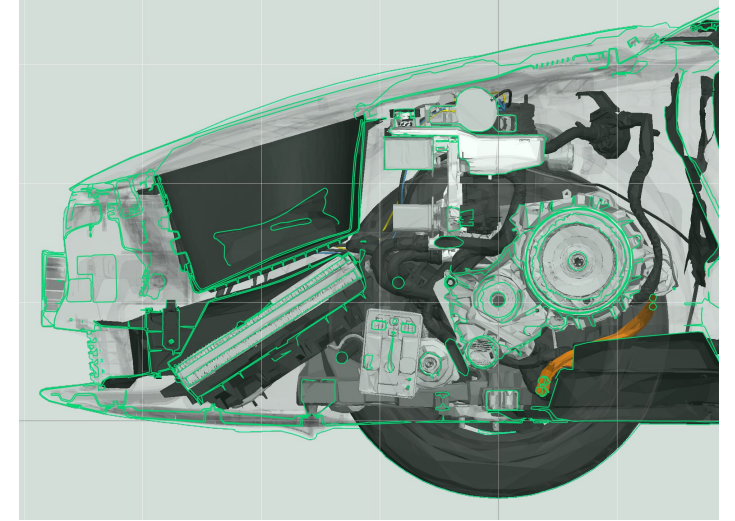
A = heat transfer area

ΔT_m = average temperature difference



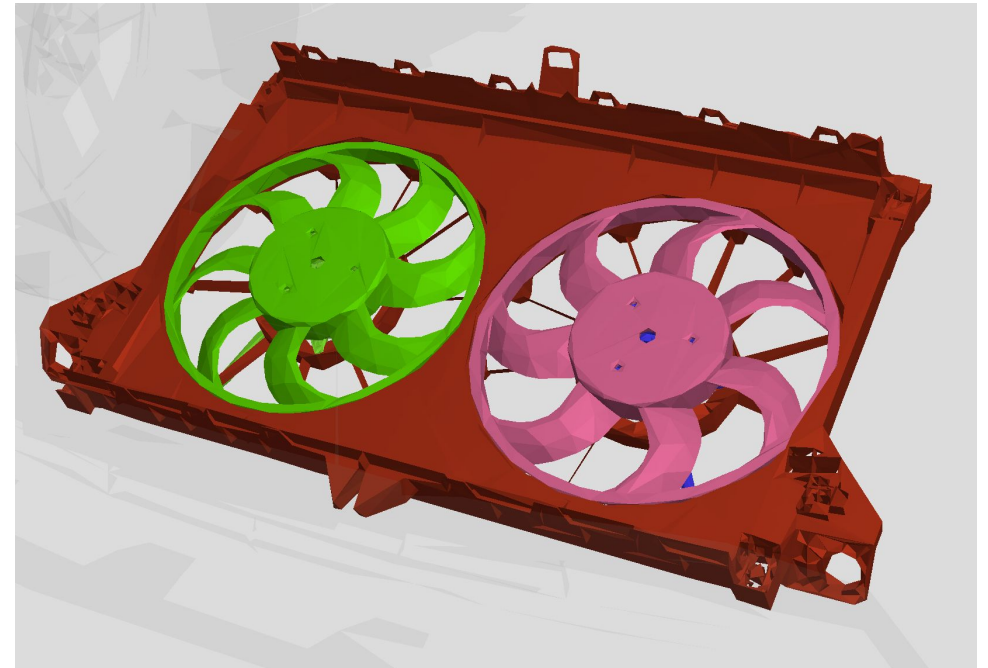
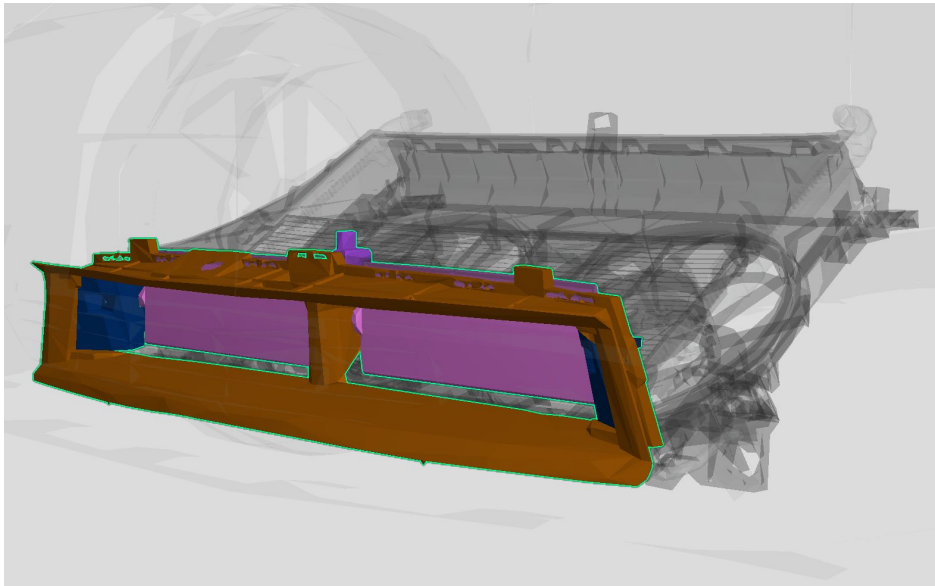
Air Guide (Duct) and Stone Guard

- Air Duct
 - Guides air from the bumper through the heat exchangers
 - Affects flow uniformity and intake pressure losses
- Stone Guard
 - Protects radiator tubes from road debris impacts
 - Optimized to improve flow uniformity



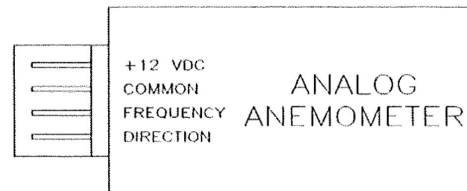
Fan System and Active Grille Shutter (AGS)

- Fan System
 - Forces airflow when ram air is insufficient (stationary vehicle)
 - Consumes electrical energy
- AGS
 - Meters flow to the heat exchangers to meet cooling demand
 - Reduces cooling drag when closed in higher speed cases

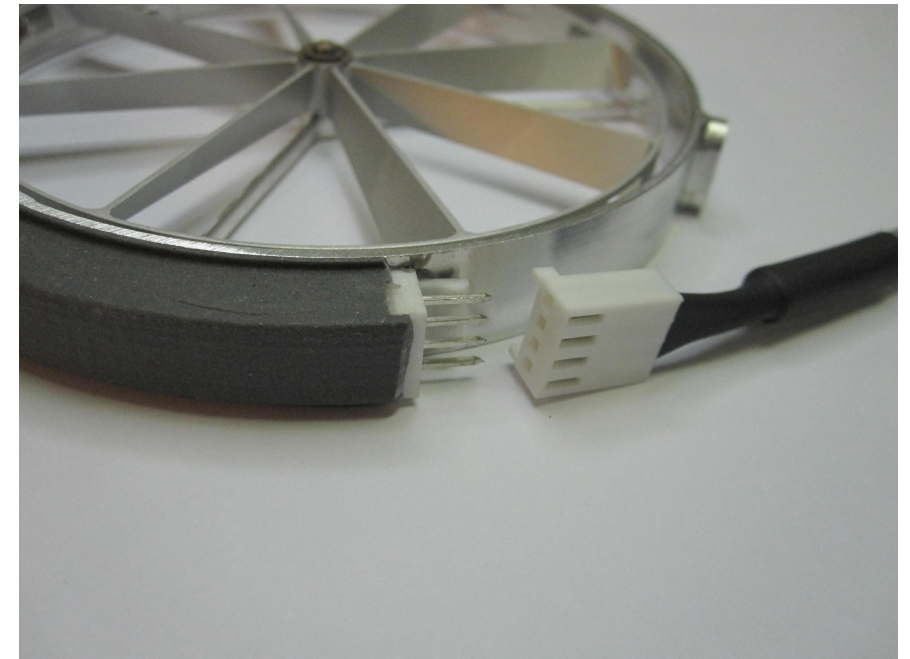


Measurement Equipment

- Vane Anemometers
 - Measure air velocity
 - Power 8-14V DC, 100mA consumption
 - Output 5V DC square wave, frequency proportional to speed
 - Output 5V DC discrete bit for flow direction
 - Temperature range: -40°C - 115°C
 - Calibrated velocity: 1.5 – 27 m/s
 - Max velocity: 53 m/s

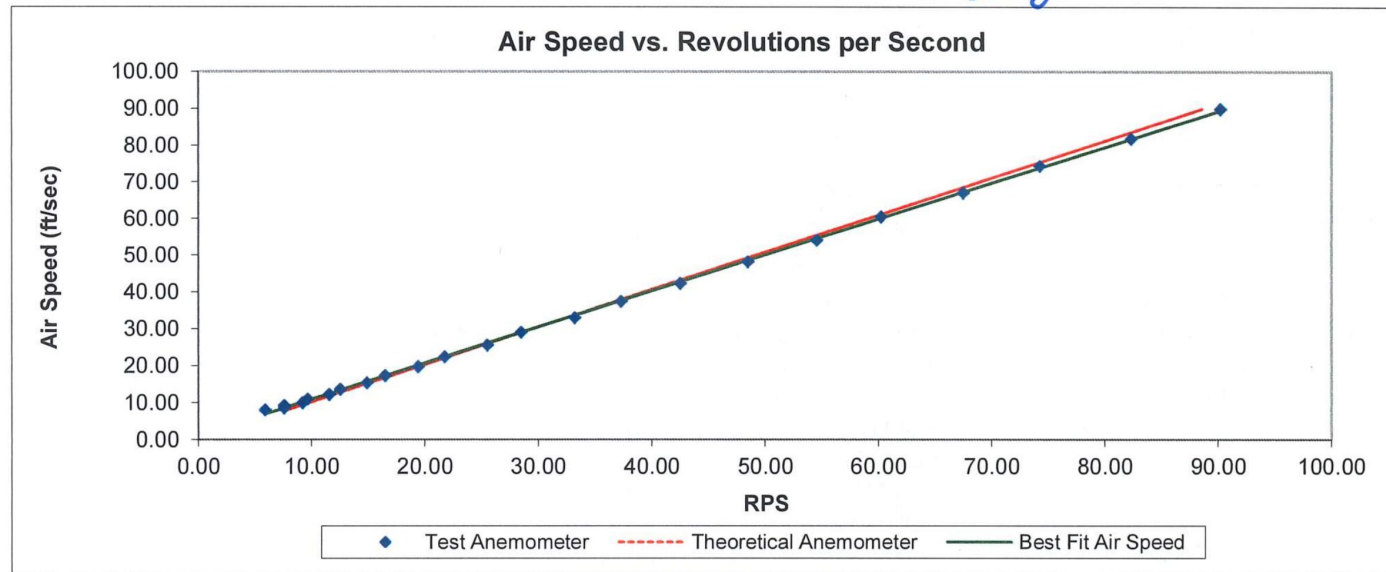


Suggested Cable	Belden 83348E
Mating Connector	Molex 22-01-3047
Mating Connector Pins (4)	Molex 08-50-0144



Anemometer Calibration

- All anemometers were calibrated
- Air velocity calculated from frequency



Calibration Constants			Linear Calibration Equation (Y = mX + b)	
Slope	Intercept	R ²	Air Velocity (Ft/Sec) = Slope * Test Revolutions (RPS) + Intercept	
0.979983	1.123307	0.999707	Test Revolutions (RPS) = Frequency (Hz) / Pulses Per Rev. (PPR)	
0.081665	1.123307		Air Velocity (Ft/Sec) = Slope * Frequency (Hz) + Intercept	

Measurement Equipment

- National Instruments cDAQ
 - cDAQ-9189 Chassis
 - NI-9361
 - Input module for frequency measurements
 - NI-9206
 - Voltage input module for directionality
 - NI-9860
 - LIN control module for fans and AGS motor



Model: cDAQ-9189

Part Number: 785065-01

[Datasheet & Specifications](#)



Model: NI-9361

Part Number: 783407-01

[Datasheet & Specifications](#)



Model: NI-9206

Part Number: 779526-01

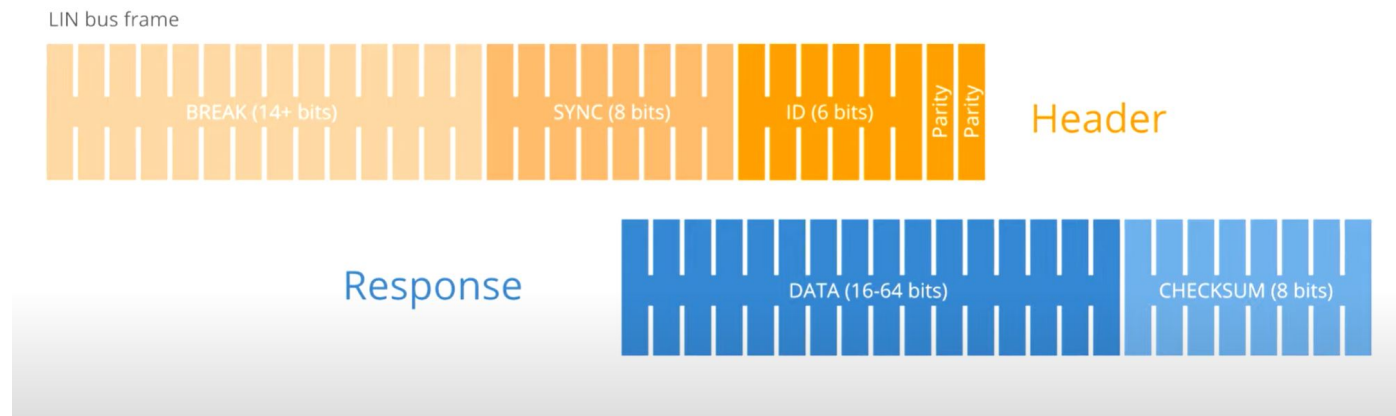


NI-9860

C Series Vehicle Multiprotocol Interface Module

Local Interconnect Network (LIN)

- Network protocol for device communication in vehicles, standardized in 1999
- Fan motors and AGS motor are controlled via LIN
- LIN cluster consists of 1 master and up to 16 slave nodes
- Single wire at 12V, 1-20Kbps
- LIN message frames include a break, sync, node ID, and the slave node response with data and a checksum
- LIN devices are defined by their LIN description file (LDF)

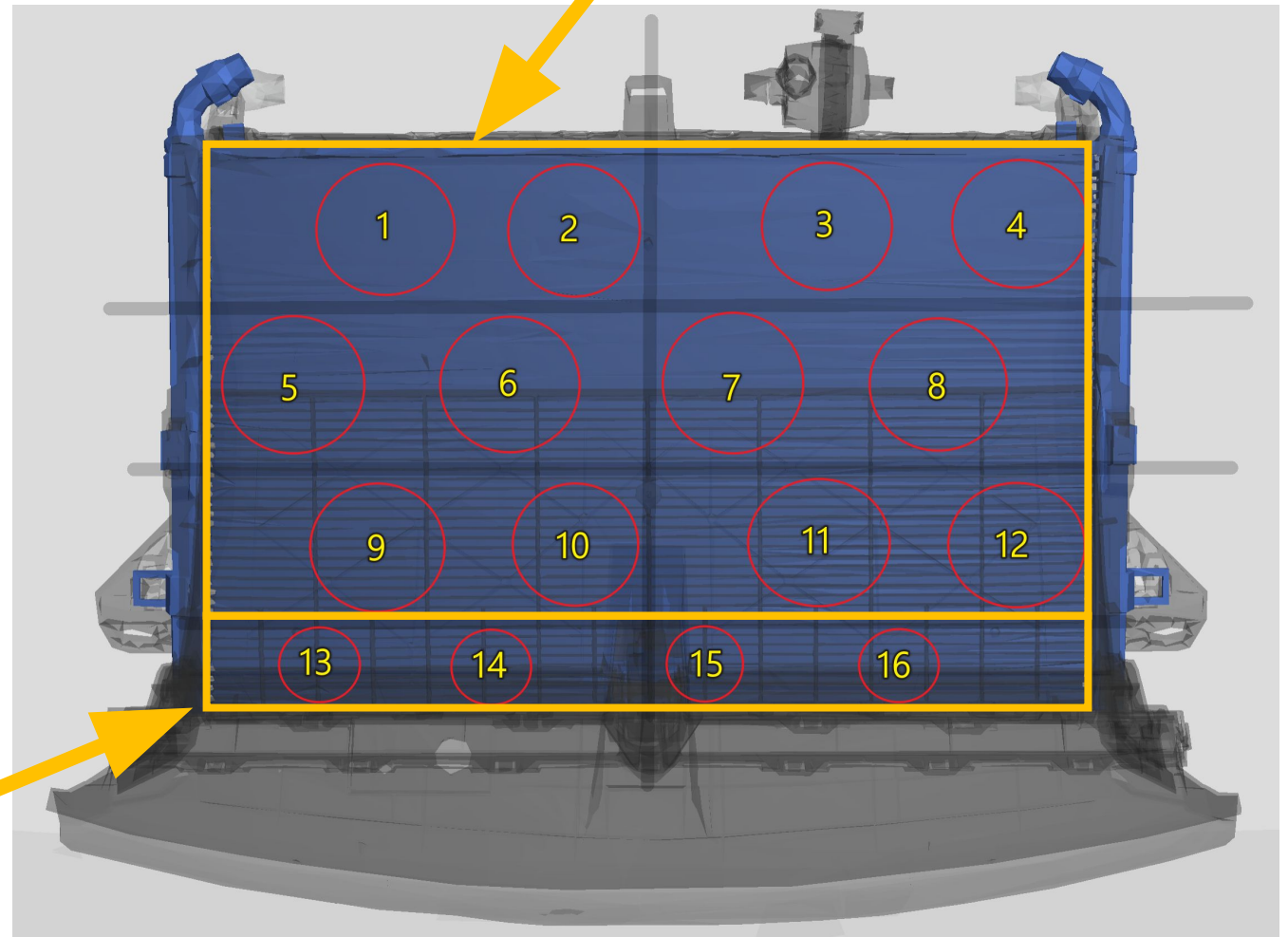


Anemometer Layout

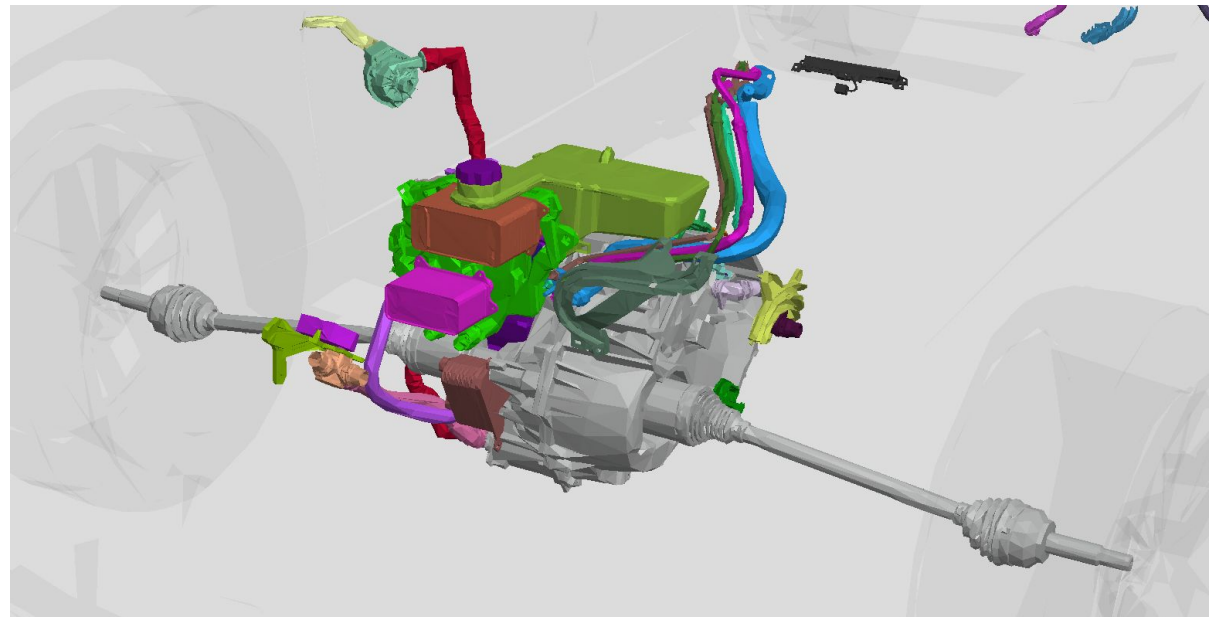
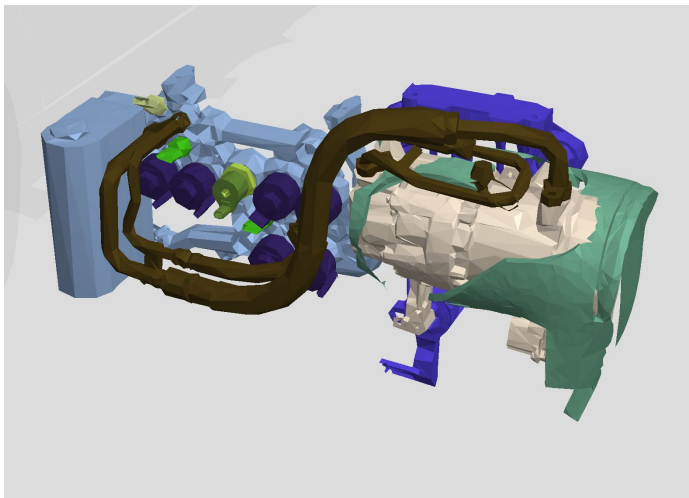
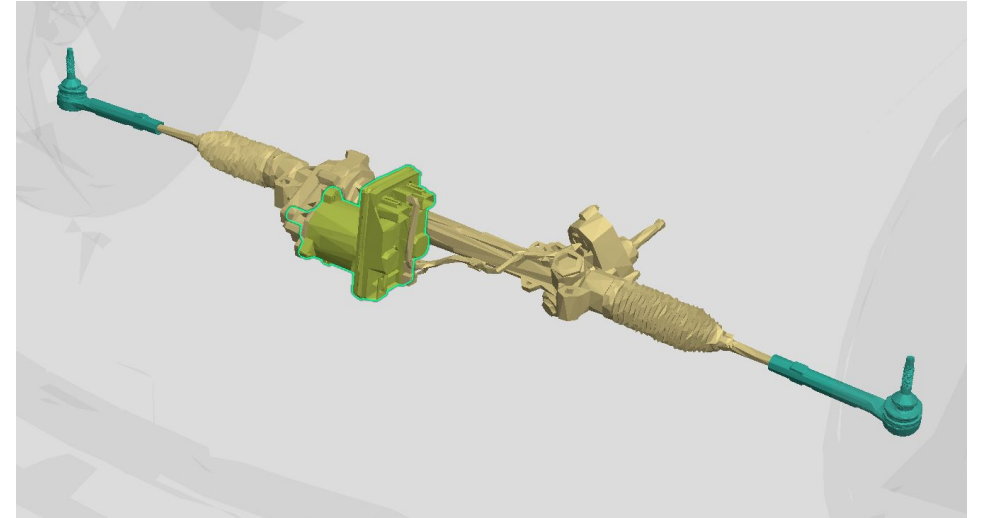
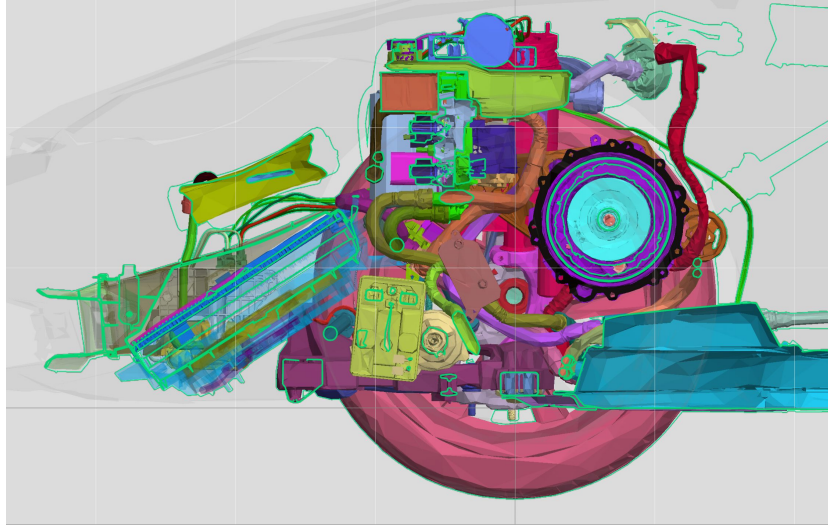
- Sixteen (16) vane anemometers affixed to the face of the radiator and condenser
- Anemometer placement determined from CFD results
- Overall flow measurement expected to match 99.7% to CFD results based on this layout
- Main radiator area measured with twelve (12) 3.5" dia anemometers
- Condenser subcooling region measured with four (4) 2" dia anemometers

Subcooling
Region

Main
Radiator
Face



Component Foam Models



Airflow Measurements

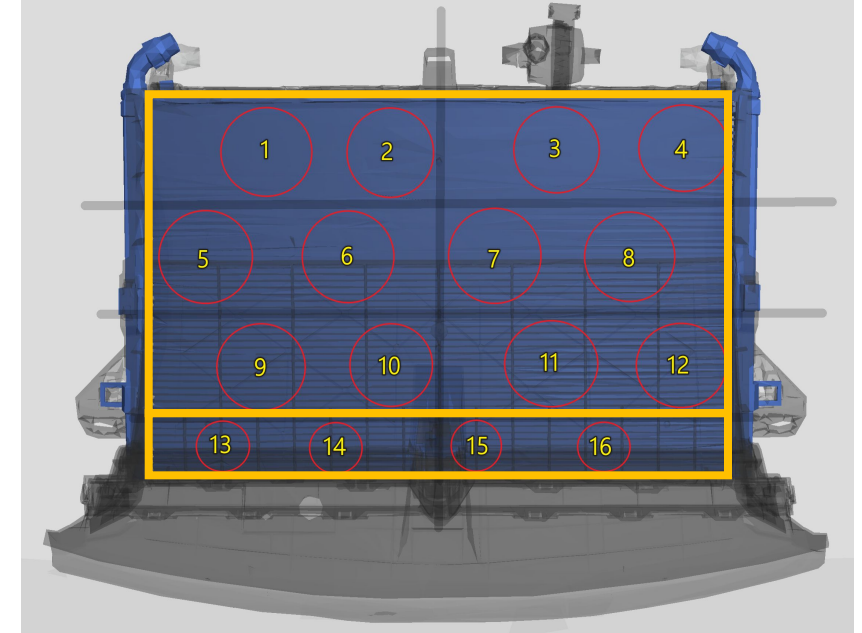
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$$Q = vA$$

Q = volume flow rate

v = flow velocity

A = flow area



- Anemometer velocities were averaged over two regions: the main radiator face, and condenser subcooling region
- The average velocity was multiplied by the area of the main radiator face, and the subcooling region separately, and these flow rates were summed to provide total flow measurement

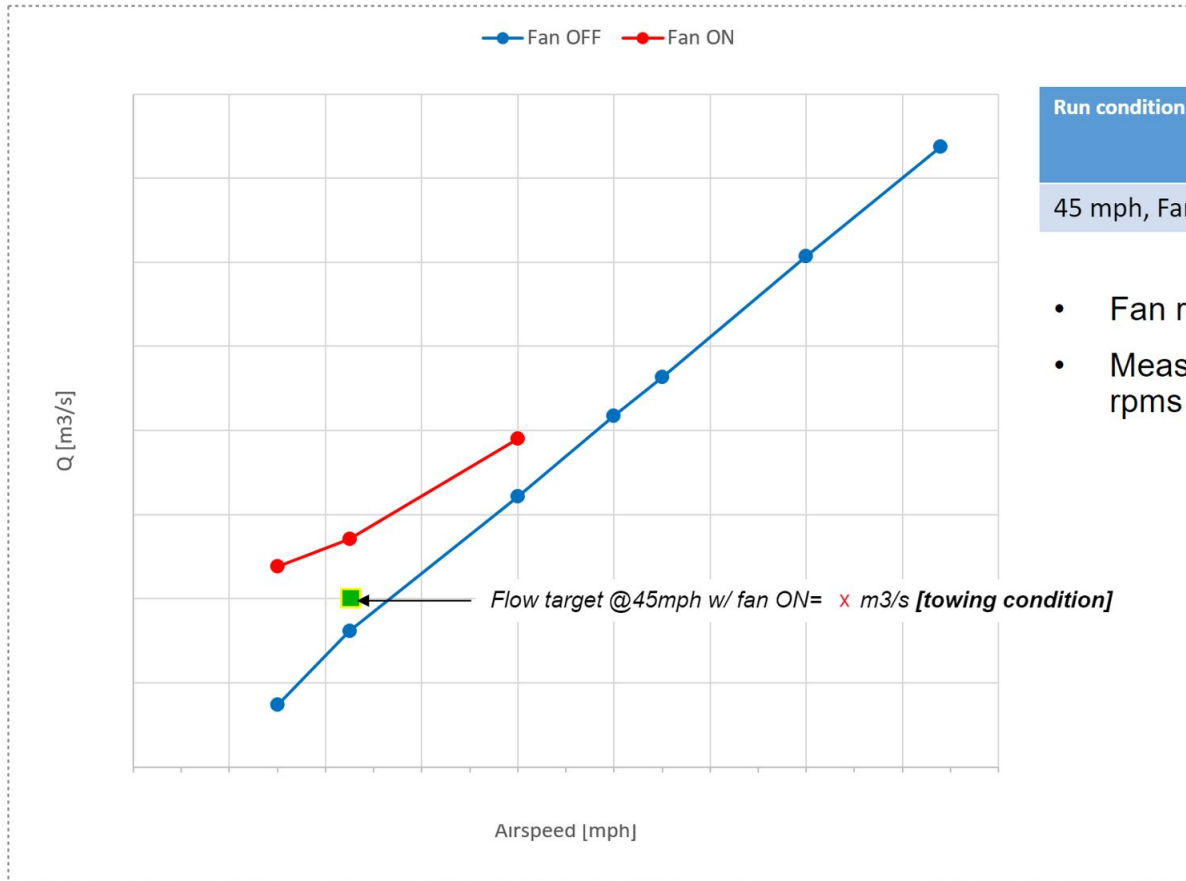
LabView and Data Logging

- LabView Program
 - Control fan and AGS motors
 - Display anemometer frequency, flow direction, and calculated flow
 - Display feedback from fan motors (actual speed, voltage, current)
- Data sampled once per second, logged with a timestamp to a .dat file continually recording while the program runs
- Button to write the next 20 seconds of data to a new file, used for recording individual test run data

Test Points

Test Number	Vehicle Speed (mph)	Fan Speed (%)	AGS Position (%)	Notes
1	80	0	100	AGS taped closed
2	80	0	100	AGS untaped
3	30, 45, 80, 100	0	0	AGS open, speed sweep fans off
4	30, 45, 80	100	0	AGS open, speed sweep fans on
5	80	0	0 – 100	AGS sweep
6	80	0	0	Yaw Sweep
7	30, 45, 80	100	0	Fascia change, speed sweep

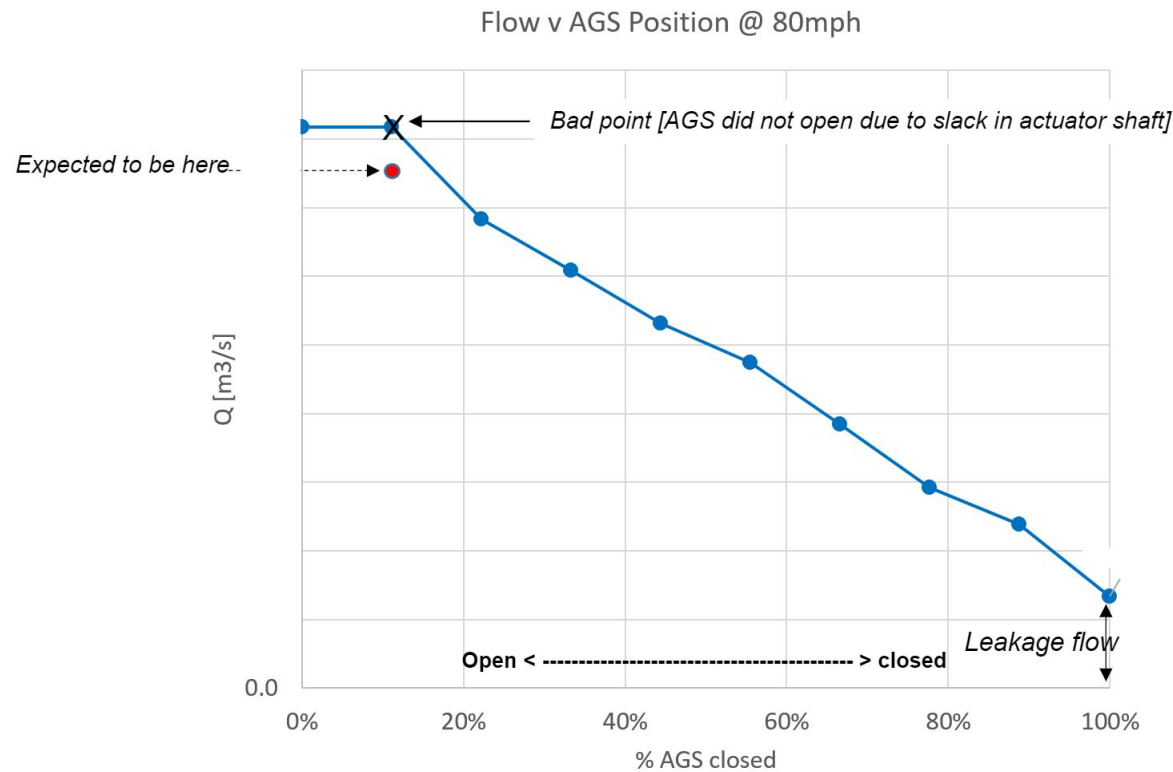
Results – Speed Sweeps



Run condition	Total cooling flow [m3/s]	Cooling flow target [m3/s]	% above target
45 mph, Fan ON [towing]	x	x	+35%

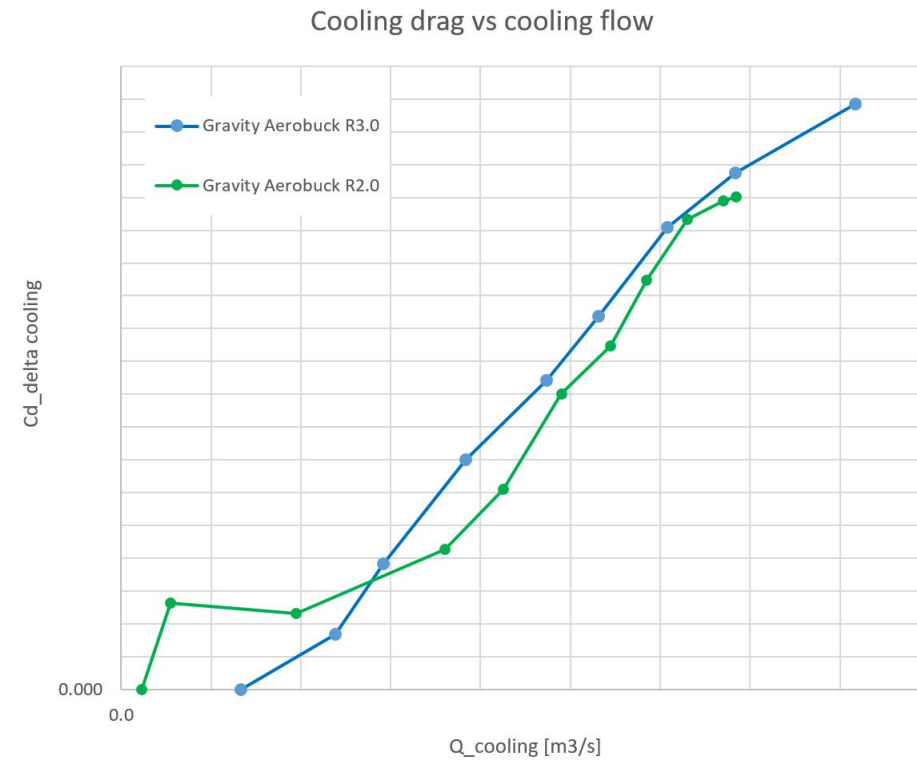
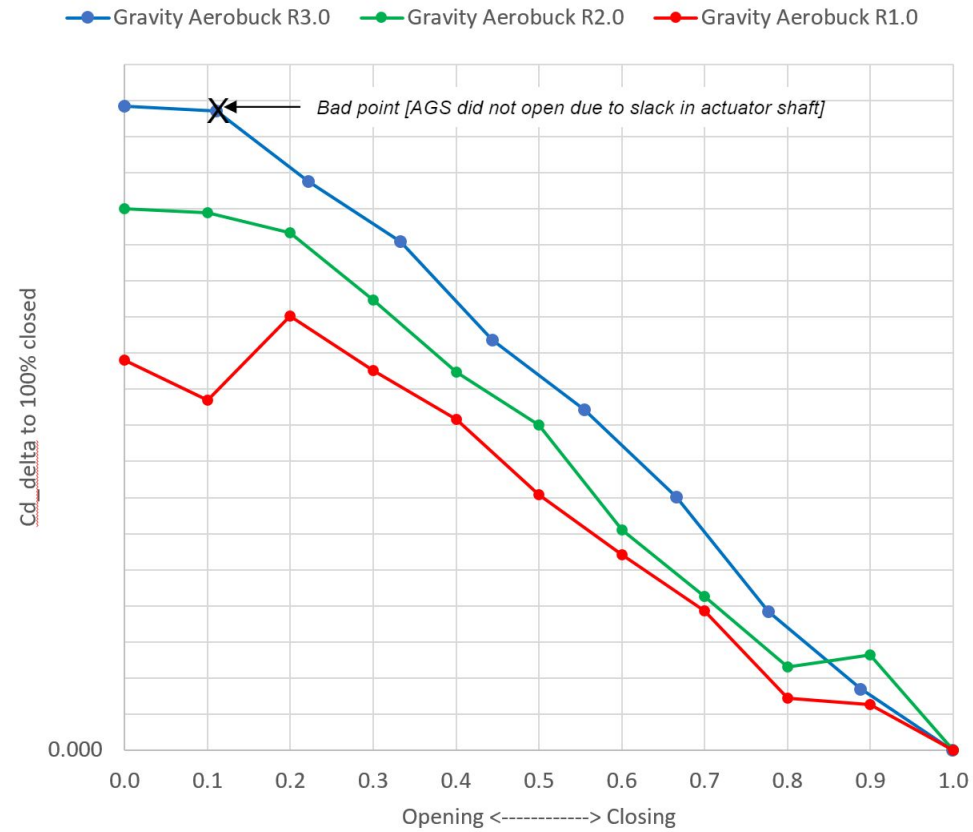
- Fan running at max rpm: x/x [DR/PS]
- Measurements indicate fans can operate at lower rpms during towing if flow required $\leq$ x m³/s

Results – AGS Sweep

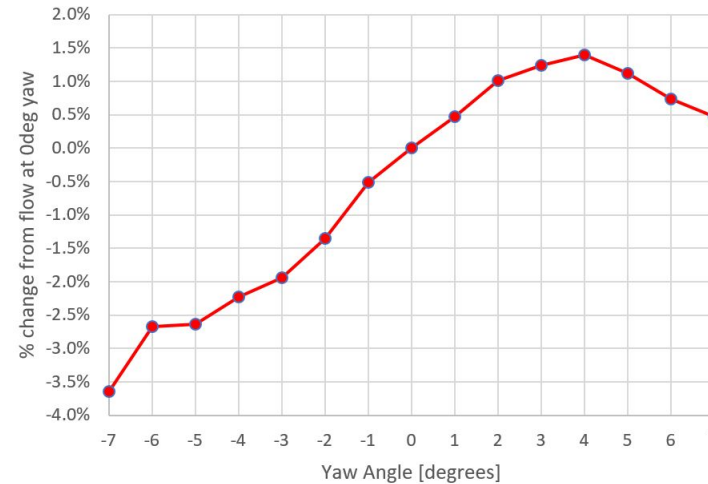
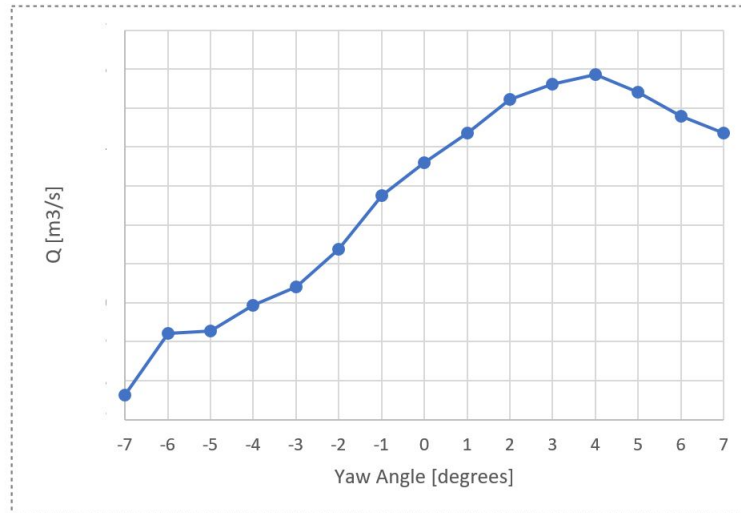


- Flow varies linearly with AGS opening %
- AGS opening was increased by 10 degrees in production intent to get more flow. Going >10 deg would have reduced subcooling flow significantly
- There was a slack between AGS actuator and blade shaft that resulted in AGS not moving for first ~10 degrees when starting from fully open or fully closed position.

Results – Cooling Drag

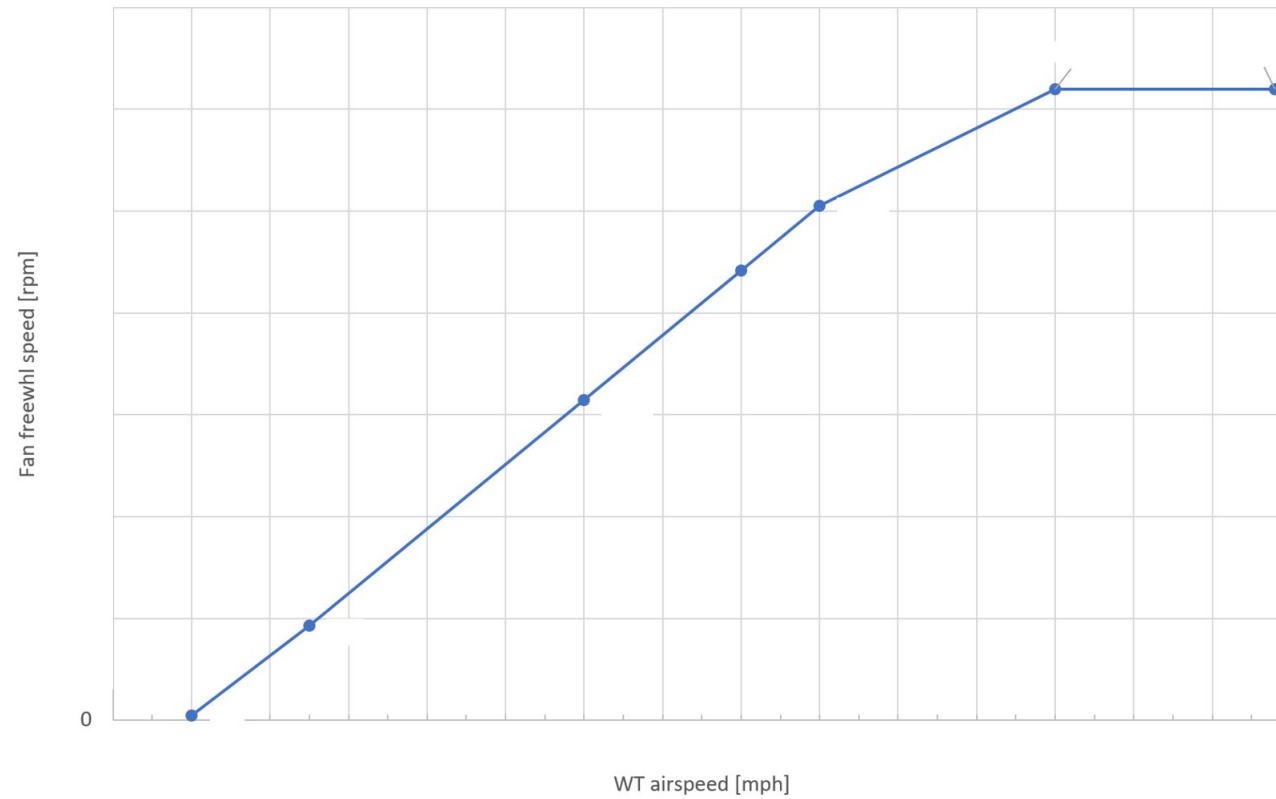


Results – Yaw Sweep



- Cooling flow varies from -3.5% to +1.5% in yaw cases compared to flow at straight line (0 deg yaw).
- Possible causes for yaw variation in flow rates
 - Flow asymmetry due to step at center post between fascia grille and AGS frame.
 - Measurements might be sensitive to anemometer arrangement

Results – Fan Freewheeling Speed



Next Steps!

- Data was used to confirm flow targets for limiting cases
- Individual anemometer measurements allowed plotting of a flow distribution to study uniformity
- Fascia design change was studied
- Follow up testing included additional stationary vehicle testing
- For future testing, more anemometers can improve the accuracy of the measurements
- Aerodynamic and cooling flow data presented to upper management
- Data and methods documented and reported internally for any future testing